

Lab 1: Build Your First Reconciliation Agent

Build a low-code reconciliation investigator

FORMAT	ESTIMATED TIME	PLATFORM
Low-code / no-code	50 minutes	Langflow

Lab premise

Build a reconciliation investigator in Langflow, connect it to Finance and Sales data, add a reviewer, and run the investigation end to end.

What You Will Learn

By the end of Lab 1, you will have built a complete Langflow agent that can investigate a reporting discrepancy using two data sources and send its draft through a reviewer before returning a final response.

Business Scenario

Morrow Peak Outfitters (MPO) is a mid-market omnichannel retailer that sells outdoor and lifestyle products. Every quarter, Finance and Sales prepare their own revenue reports for leadership. The reports are supposed to describe the same business performance, but recurring discrepancies have turned each reconciliation cycle into a frustrating back-and-forth across Finance, Sales, and the data team.

Finance quarterly report	\$4,200,000
Sales quarterly business review (QBR)	\$3,800,000
Difference between the two reports	\$400,000

At the end of Q3, it happened again. Finance reports \$4.2 million in net revenue for the U.S. Direct business, while Sales reports \$3.8 million. Both teams believe they are looking at the same business, the same quarter, and data from the same underlying warehouse, but no one can immediately explain the \$400,000 gap. The discrepancy is now visible at the executive level, and leadership needs a clear, evidence-supported explanation before the numbers are used in the next review.

What the Team Wants From the Agent

Today, a human analyst has to move between reporting views, compare how transactions are handled, read business definitions, trace changes, and follow up with other teams before they can explain a mismatch. That work can take hours or days when the cause is not obvious. After repeated rounds of manual investigation, MPO wants to use an agent to speed up the first pass: inspect the available sources, decide what evidence to look at, compare what it finds, and package the evidence for an analyst.

The agent is not the final decision-maker. A human analyst reviews the evidence, decides whether the explanation is sound, and is the last line of defense before anything is presented to leadership. In Lab 1, the agent starts with two sources only: the Finance Q3 extract and the Sales Q3 extract.

Business objective

Reduce the time it takes to understand reporting discrepancies so experienced analysts can spend less time gathering evidence and more time validating the explanation and giving leadership the right answer.

Files for This Lab

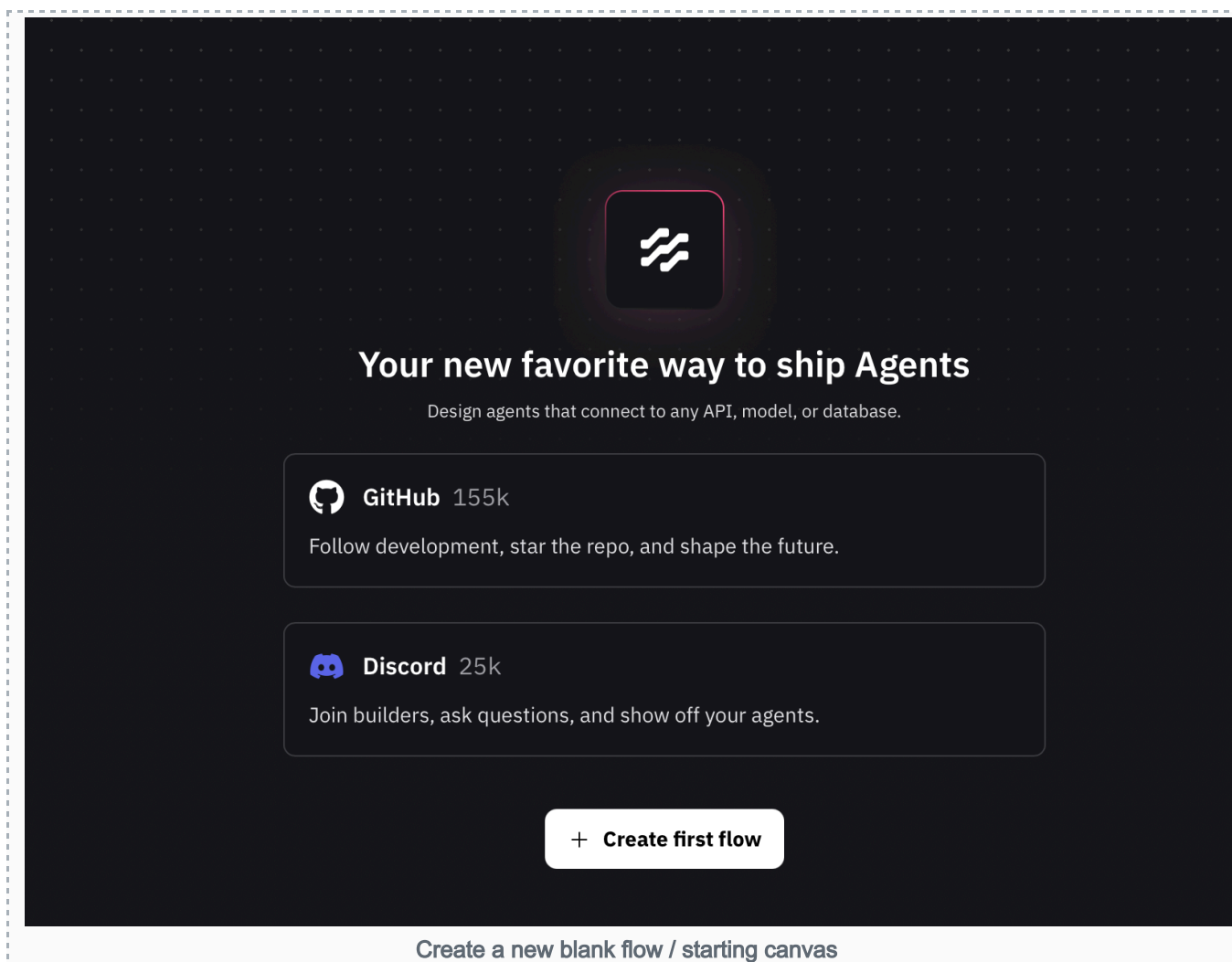
- data/morrow_peak_finance_q3_reconciliation_extract_v3.csv
- data/morrow_peak_sales_q3_reconciliation_extract_v3.csv

Each file has a REPORT_SUMMARY row with the official Q3 net revenue and additional detail rows. REVENUE_SAMPLE contains partial activity; the refund rows are the complete comparison set for this lab. Use the summary row for the reported total and the detail rows for supporting evidence.

Part 1 — Create and Configure the Flow

Step 1 Open Langflow and create a new flow

Open Langflow and create a new blank flow. If Langflow offers starter templates, choose the blank/new-flow option. You will add each component shown in the target architecture yourself.



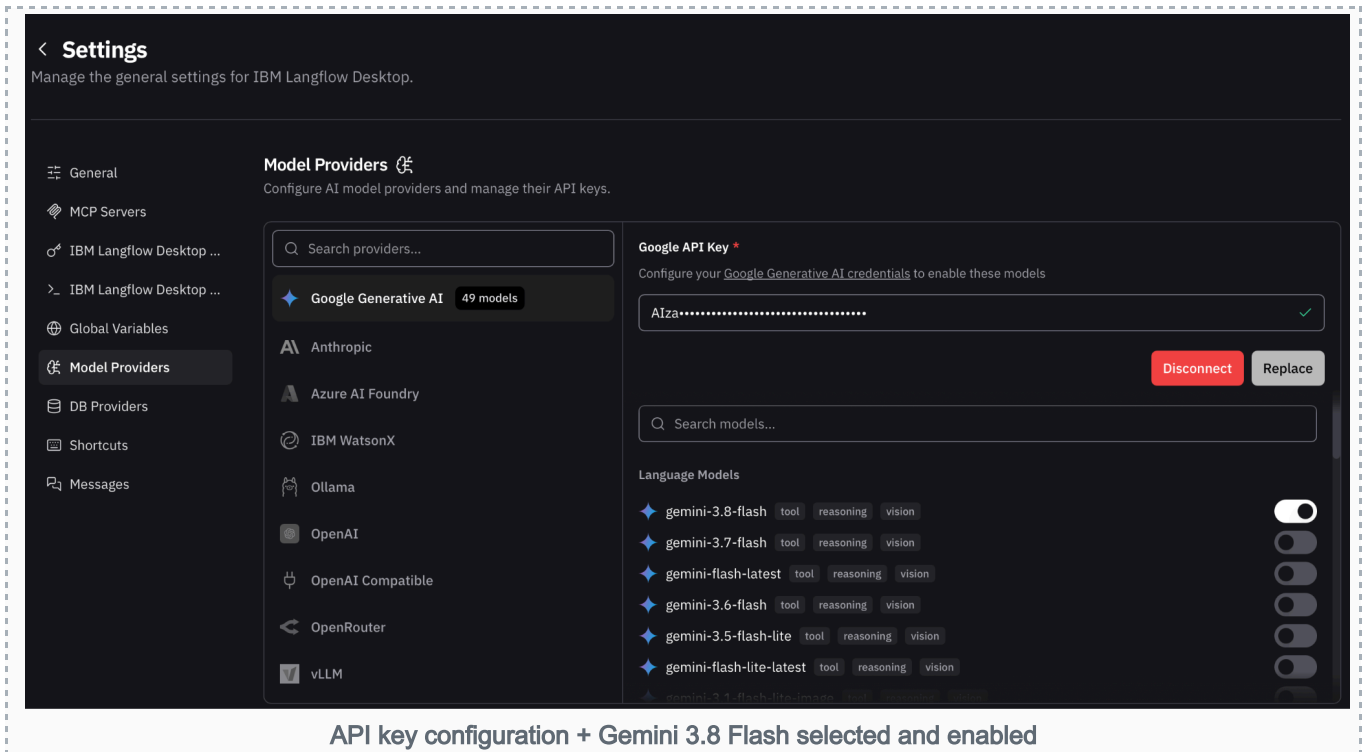
Step 2 Configure the Google Gemini API key and model

When Langflow prompts you to configure the model provider, use the API key supplied by the instructor. Select the Google Gemini provider and choose the model used for this lab:

Model

gemini-3.8-flash

Enable/toggle the model so it is available to the components in your flow. You will use the same model for the Reconciliation Investigator and the Reflection / Reviewer.



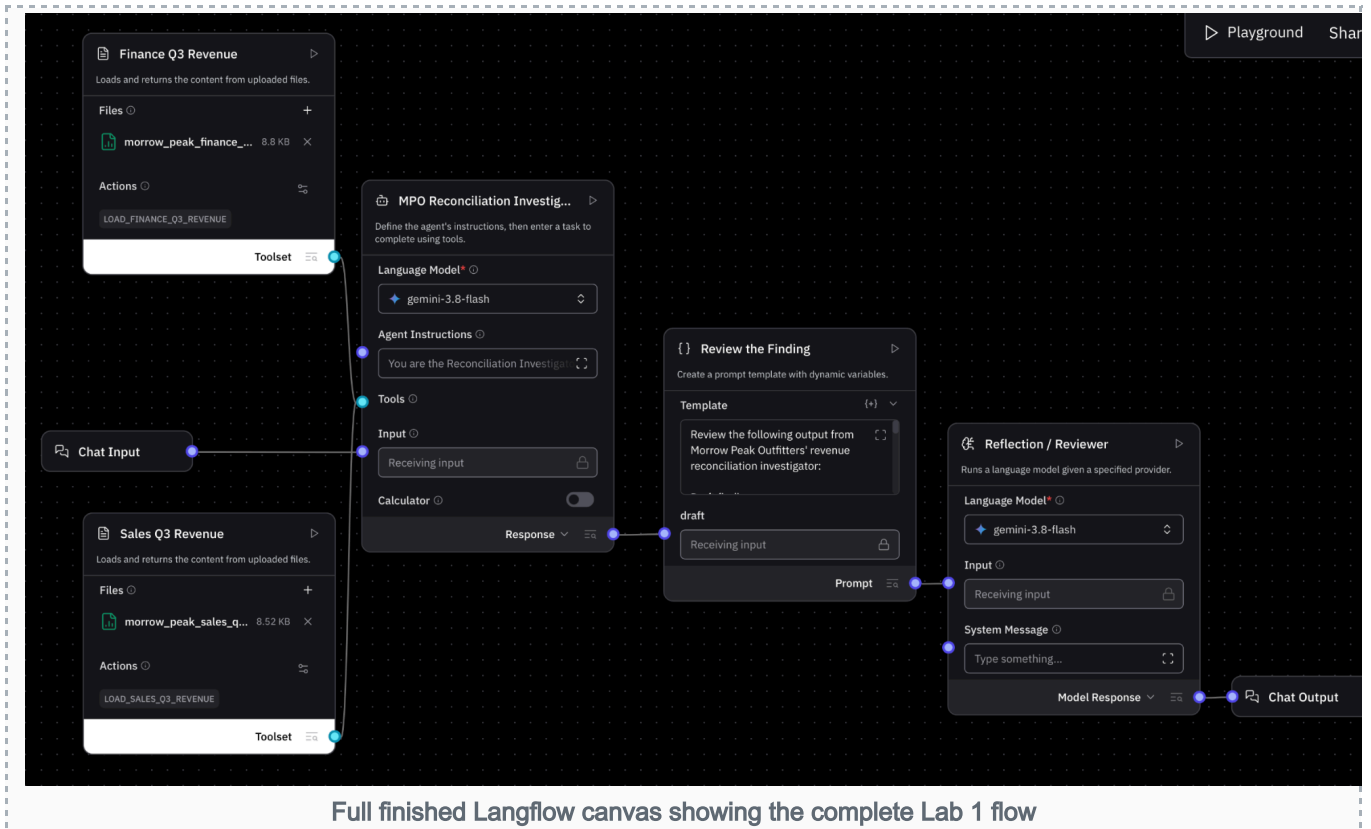
API key note

Use only the temporary workshop credential supplied by the instructor. Do not paste a personal production credential into a shared screen, chat, or lab document.

Step 3 Review the architecture before you build

Your finished flow will contain one agent, two data tools, one reflection prompt, one reviewer model, and one chat output. The reviewer is a standard language-model call, not a second agent.

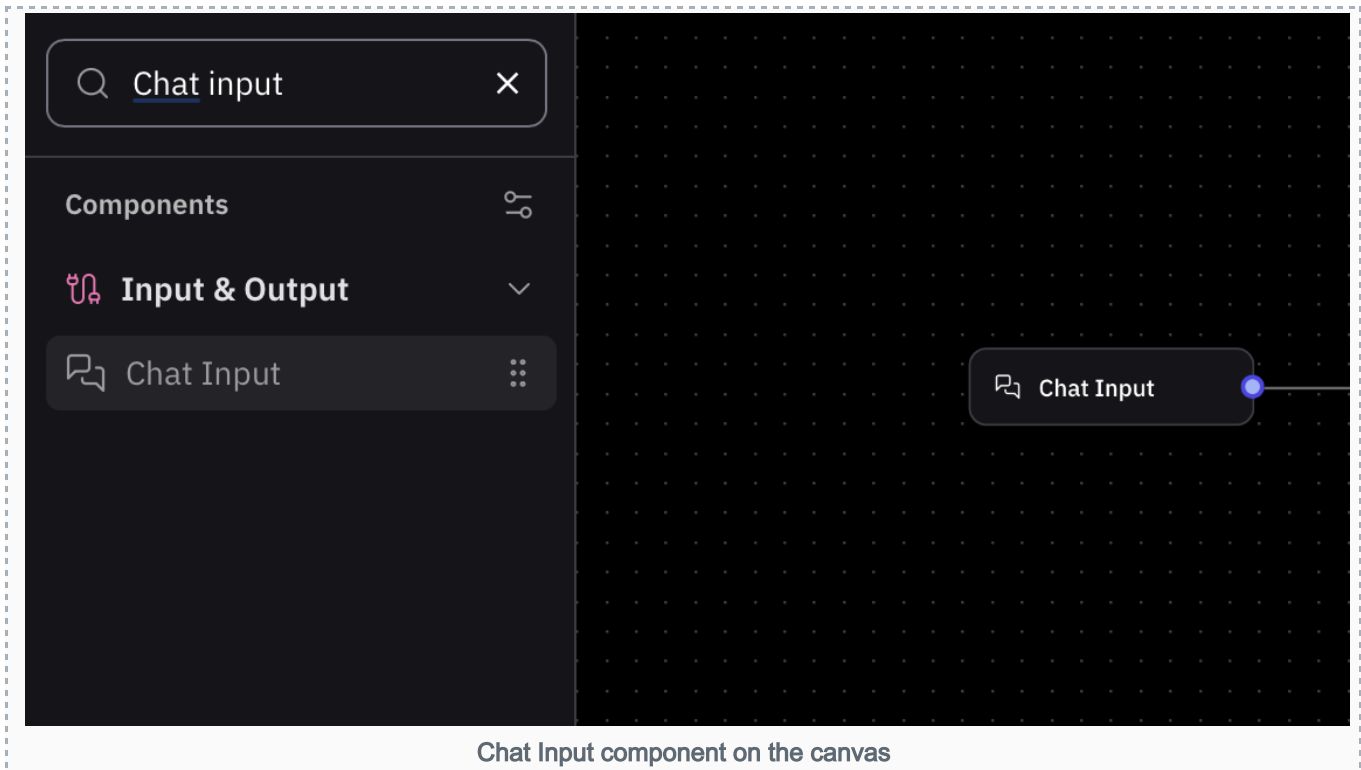
Target architecture



Part 2 — Build the Agent and Its Tools

Step 4 Add the Chat Input component

Add a Chat Input component to the canvas. This is where the user's question enters the flow. Keep the runtime question here; the Agent Instructions will hold the agent's standing role and rules.



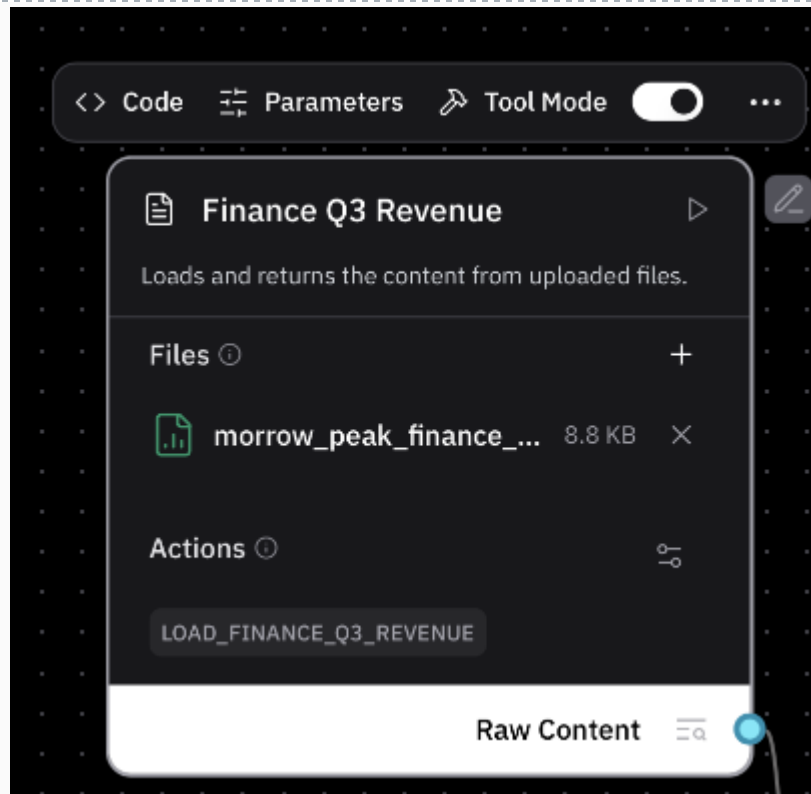
Step 5 Add and configure the Finance reporting tool

Add a Read File component and rename it Finance Q3 Revenue. Upload the Finance v3 CSV. Configure the Read File component so the Agent can use it as a tool by toggling on “tool mode”.

In the component's Actions configuration, use the following slug name and description. The Agent sees the tool name, description, and schema when deciding which tool to call.

Slug name: load_finance_q3_revenue

Description: Loads the Finance Q3 revenue extract. Use this tool when you need Finance-reported Q3 revenue, Finance refund treatment, or evidence from the Finance reporting source.



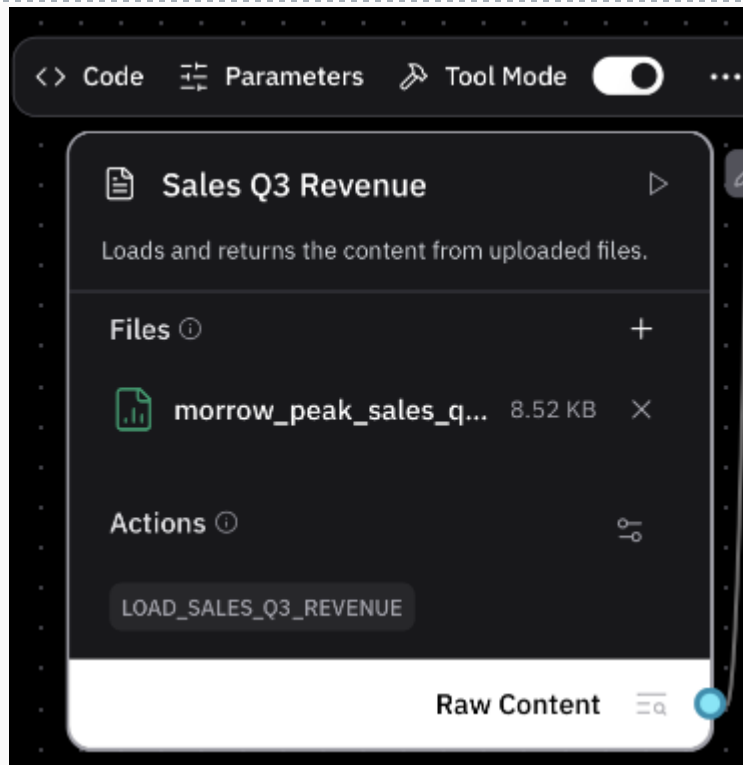
Finance Read File component with file loaded and action/tool configuration visible

Step 6 Add and configure the Sales reporting tool

Add a second Read File component and rename it Sales Q3 Revenue. Upload the Sales v3 CSV and expose it as a tool by toggling on “tool mode”. Use the following name and description exactly.

Slug name: load_sales_q3_revenue

Description: Loads the Sales Q3 revenue extract. Use this tool when you need Sales-reported Q3 revenue, Sales refund treatment, or evidence from the Sales reporting source.

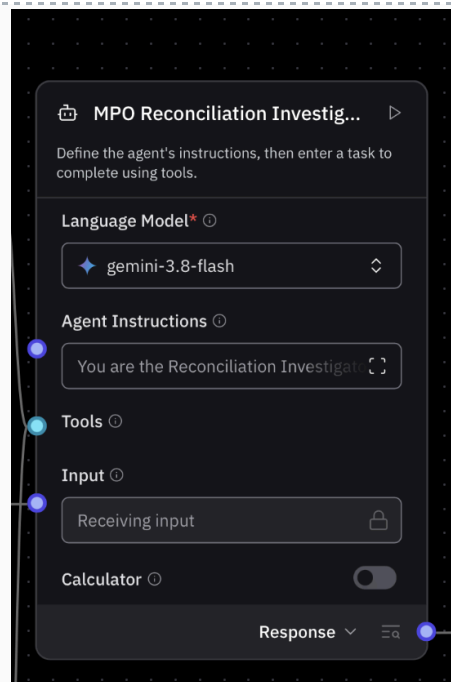


Sales Read File component with file loaded and action/tool configuration visible

Step 7 Add the Reconciliation Investigator Agent

Add an Agent component and rename it MPO Reconciliation Investigator. Configure it to use *gemini-3.8-flash* and the instructor-provided Gemini credential.

Connect the Chat Input to the Agent input. Connect both Read File Toolset outputs to the Agent Tools input.



Agent connected to Chat Input plus both Finance and Sales tool outputs

Paste the following into the Agent Instructions field:

You are the Reconciliation Investigator for Morrow Peak Outfitters.

Your role is to investigate reporting discrepancies using the available reporting sources and provide concise, evidence-supported findings to business stakeholders.

When investigating a discrepancy between Finance and Sales:

- Use both the Finance and Sales reporting sources.
- Call each reporting source no more than once during the investigation.
- Each source contains a REPORT_SUMMARY record. Use reported_q3_net_revenue_usd from that record as the team's official reported Q3 net revenue.
- Do not recompute the official reported revenue from detailed records.
- Records with record_type = REVENUE_SAMPLE are a partial sample of revenue activity, not the complete reporting population. Do not aggregate these rows to infer the full revenue baseline or use differences in sample subtotals as a cause of the reporting discrepancy.
- Use detailed records to look for specific, comparable evidence that may explain why the reported figures differ.
- Compare relevant dates, inclusion behavior, shared identifiers, and amounts when useful.
- Base every factual claim on evidence returned by the available tools.
- Do not invent missing information or infer unsupported causes.
- Do not modify source data.

Once you identify one specific, evidence-supported cause and the amount associated with it, stop investigating and provide your finding. Do not continue searching for additional causes after you have found a supported explanation.

Your final finding should include the Finance and Sales reported figures, the cause you identified, the amount associated with that cause, and a brief explanation of the supporting evidence.

Keep the response concise and written for a business stakeholder.

Step 8 Explicitly disable the Agent's built-in Calculator

In this Langflow Agent, the built-in Calculator can be enabled even when no Calculator component appears on the canvas. Add the Calculator option in the Agent's parameter settings so its toggle is visible, then turn it OFF. Leave it off for this lab; when enabled, it can repeatedly call evaluate_expression and hit the recursion limit.

Component Parameters

Adjust component parameter visibility and define API inputs.

Language Model

Default: Empty

API

Remove

Agent Instructions

Default: You are a Langflo...

API

Remove

Tools

Default: Empty

API

Remove

Input

Default: Empty

API

Remove

Calculator

Default: Enabled

API

Remove

API Key

Default: Empty

API

+ Add

Context ID

Default: Empty

API

+ Add

Number of Chat Hi...

Default: 100

API

+ Add

Max Tokens

Default: 0

API

+ Add

Agent settings showing the Calculator control exposed and toggled OFF

Part 3 — Add Reflection and the Final Output

Step 9 Add the Reflection Prompt

Add a Prompt component and place it after the Agent. Set the Agent response as a prompt variable named draft. The reviewer will receive the completed prompt instead of the Agent message directly.

Reflection Prompt template

Review the following output from Morrow Peak Outfitters' revenue reconciliation investigator:

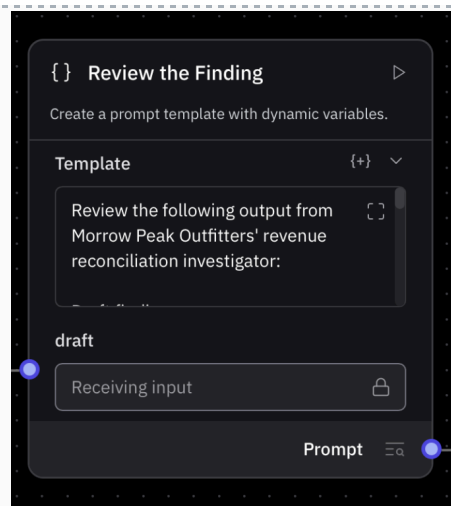
Draft finding:
{draft}

Evaluate the finding against these criteria:

- It identifies a specific cause for the reporting difference.
- The cause is supported by evidence in the response.
- It states the relevant amount.
- It does not make unsupported claims.
- The explanation would be understandable to a business stakeholder.

If the criteria are satisfied, return the finding as a polished final response.

If they are not satisfied, improve the response using only the evidence contained in the draft. Do not introduce new facts.



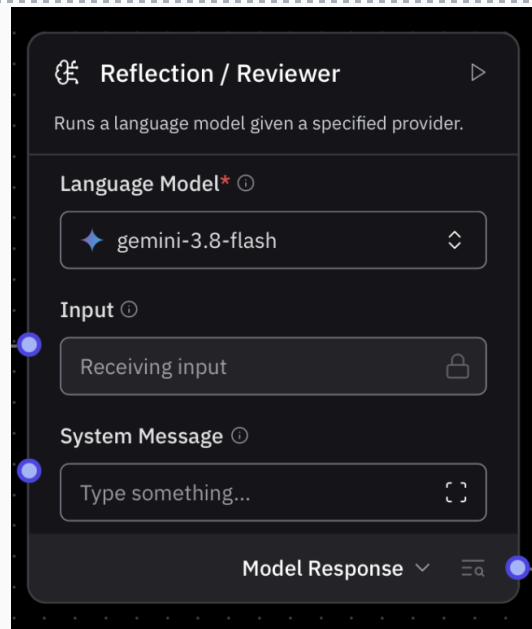
Reflection Prompt with {draft} variable configured and connected to Agent response

Step 10 Add the Reflection / Reviewer model

Add a Language Model component and rename it Reflection / Reviewer. Configure it with the same Gemini model used by the Agent: gemini-3.8-flash.

Connect the Reflection Prompt output to the Reviewer Input. In the Reviewer System Message, use:

You are a careful reviewer of business reconciliation findings.



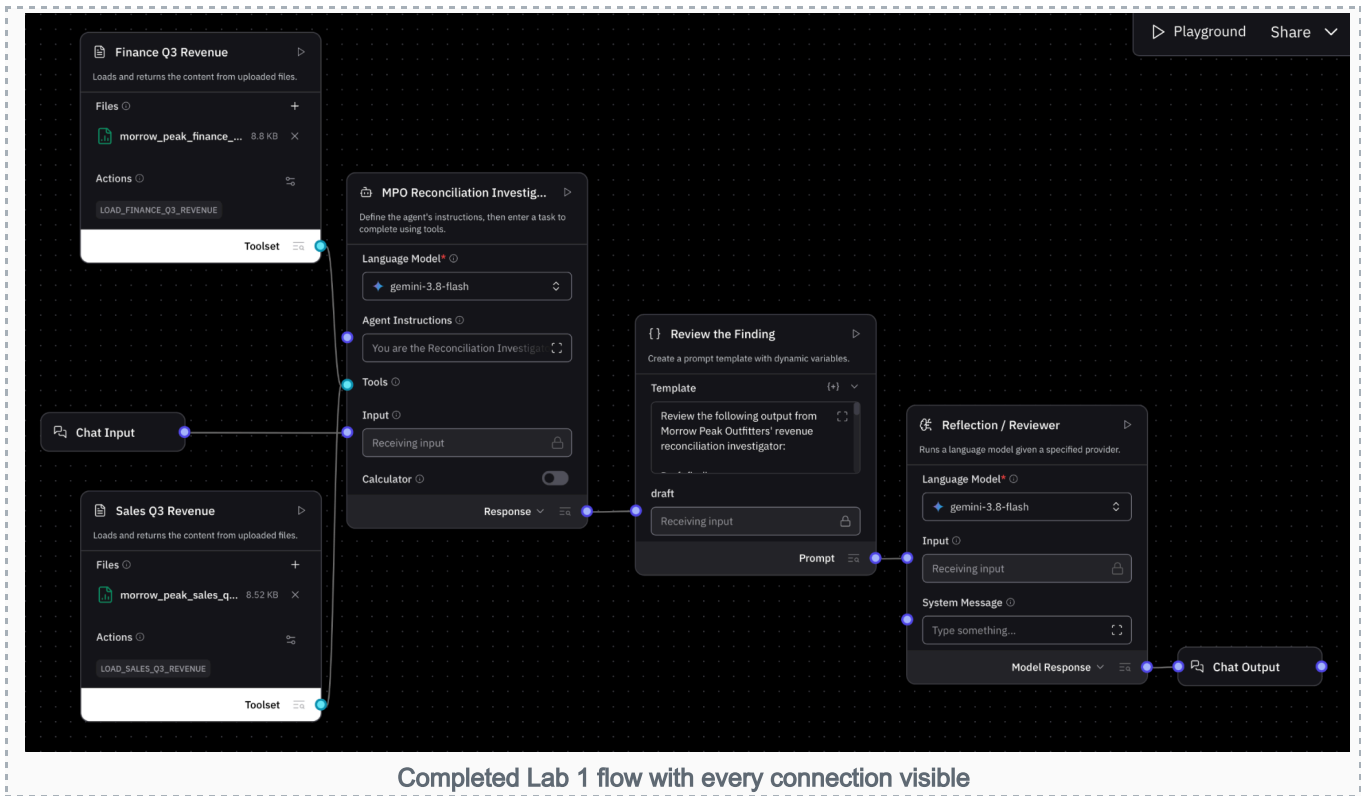
Reflection / Reviewer model with Gemini 3.8 Flash selected and prompt connected to Input

Step 11 Add Chat Output and complete the wiring

Add a Chat Output component and connect the Reviewer response to it. Route the Agent response through the Reflection Prompt before it reaches the Reviewer; connecting the Agent directly to the Reviewer can cause Gemini message-role errors.

FROM	TO
Chat Input	MPO Reconciliation Investigator — Input
Finance Q3 Revenue — Toolset	MPO Reconciliation Investigator — Tools
Sales Q3 Revenue — Toolset	MPO Reconciliation Investigator — Tools
MPO Reconciliation Investigator — Response	Reflection Prompt — draft
Reflection Prompt — Output	Reflection / Reviewer — Input
Reflection / Reviewer — Response	Chat Output — Input

Port labels may vary slightly by Langflow version; follow the data direction shown above.



Completed Lab 1 flow with every connection visible

Part 4 — Run the Investigation

Step 12 Do a final pre-run check

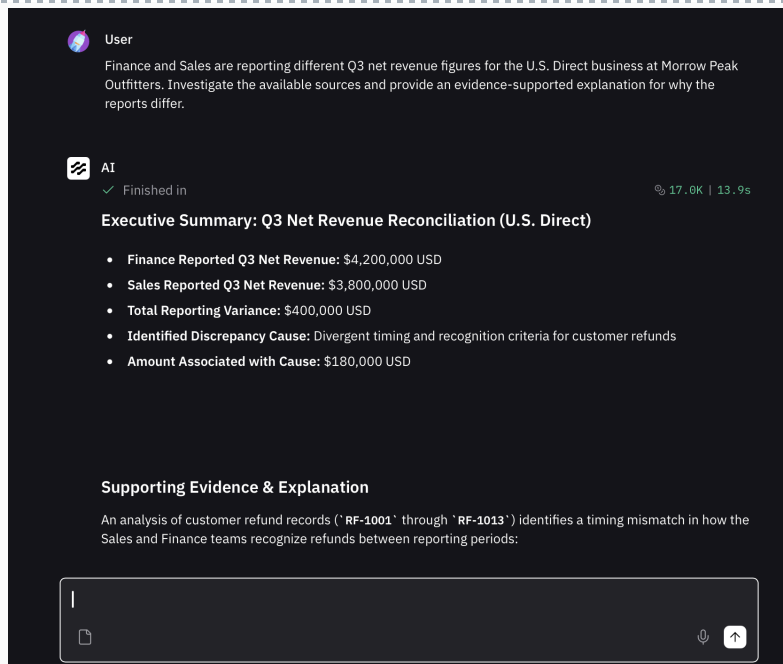
- Finance v3 and Sales v3 CSVs are loaded into the correct Read File components.
- Finance and Sales tools have distinct action names and descriptions.
- Both data tools are connected to the Agent Tools input.
- The Agent and Reviewer both use gemini-3.8-flash.
- The Agent's built-in Calculator is OFF.
- Agent output flows through the Reflection Prompt before reaching the Reviewer.
- You are starting a fresh Playground conversation for the final run.

Step 13 Run the business request

Open the Playground(located in top right side of screen) and submit this exact request:

Finance and Sales are reporting different Q3 net revenue figures for the U.S. Direct business at Morrow Peak Outfitters. Investigate the available sources and provide an evidence-supported explanation for why the reports differ.

Your output may look different from the person next to you. Check that the reported figures are correct, the claims are supported by the available evidence, and the final response is understandable to someone who was not part of the investigation.



Playground showing the final response after Agent → Reflection Prompt → Reviewer

STOP — Wait for the instructor debrief

Once you have a final response on screen, stop here. Keep the response visible and wait for the instructor before continuing. Do not change your flow yet.

If You Finish Early: Try One Change

After the instructor debrief, use any remaining time to change one part of the flow and rerun the same business request. Change only one thing at a time so you can see what that change does.

- **Tool description:** revise one tool description. Does the Agent choose or use the source differently?
- **Reviewer criteria:** add, remove, or tighten one review criterion. What changes in the final response?
- **Agent instruction:** change one operating instruction and rerun the same request. What changes in the investigation or final answer?

Compare the modified run with your original run. Look at the tools used, the evidence selected, the response structure, and the number of model/tool steps.

Troubleshooting

SYMPTOM	WHAT TO CHECK
Recursion limit of 35 reached	Verify the Agent Calculator is exposed and toggled OFF. Start a fresh Playground session. Do not increase the recursion limit for this lab.
Agent uses only one business source	Verify both Read File Toolset outputs are physically connected to Agent Tools and confirm the Finance/Sales action names and descriptions are distinct.
Reviewer error about a model turn / message role	Confirm the Agent response goes into the Reflection Prompt first, and the Reflection Prompt output goes into Reviewer Input. Do not wire Agent directly to Reviewer Input.
429 / quota error from Gemini	Pause and notify the instructor. Do not rotate random API keys or change the lab architecture; the workshop credential/project may have reached a provider quota.
Results look like an older run	Start a new Playground conversation after changing tools, prompts, files, or model settings.

Recovery flow

If you are still blocked after the troubleshooting checks and the instructor directs you to do so, import the completed Lab 1 reference flow provided with the workshop materials. Re-enter the workshop Gemini credential if prompted, confirm the model selection, and continue with the run. The reference flow is a recovery path so you can rejoin the lab without spending the session debugging the visual builder.

If your flow still fails after these checks, capture the Agent trace and show it to the instructor. The trace should make it possible to see which tool or model call is repeating.

Lab 1 Completion Checklist

- ☐ I created a new Langflow flow from a blank canvas.
- ☐ I configured Gemini 3.8 Flash with the workshop credential.
- ☐ I added separate Finance and Sales reporting tools with clear descriptions.
- ☐ I built one Reconciliation Investigator Agent and disabled its built-in Calculator.
- ☐ I added a Reflection Prompt and a separate Reviewer model call.
- ☐ I wired the final Reviewer response to Chat Output.
- ☐ I ran the reconciliation request in a fresh Playground session.
- ☐ I inspected the evidence used in the final response.
- ☐ I can explain how the tools, Agent instructions, Reflection Prompt, and Reviewer contributed to the result.

Before you leave Lab 1

Save or export your flow if instructed. You will use the same Morrow Peak Outfitters scenario in the next labs.